

1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Multi-center Exploratory Safety and Efficacy Study With PSMA, SSTR2 and GRPR Targeted Radioligand Therapy in Metastatic Neuroendocrine Prostate Cancer
Short Title/Acronym: NEPC Study **Protocol Number:** CAAA617H12101 / 2023-505655-43 **NCT Number:** NCT06379217 **Posting ID:** 61543512700935559 **Sponsor Name:** Novartis Pharmaceuticals
Investigational Product: Targeted Radioligand Therapies (RLT) including [177Lu]Lu-PSMA-617, [177Lu]Lu-NeoB, and [177Lu]Lu-DOTA-TATE **Phase of Development:** Phase I **Trial Status:** Recruiting
Medical Monitor: Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the change in the expression of treatment targets on the surface of tumor cells (Prostate Specific Membrane Antigen [PSMA], Somatostatin Receptor 2 [SSTR2], and Gastrin Releasing Peptide Receptor [GRPR]) between the baseline and following targeted radioligand therapy (RLT).
Secondary Objectives: To explore the safety, tolerability, radiographic disease progression, and overall survival outcomes following the assigned targeted radioligand therapy.

2.2 Background & Rationale Metastatic neuroendocrine prostate cancer (mNEPC) is an aggressive variant of prostate cancer with a poor prognosis and limited therapeutic options. This study aims to investigate whether selecting a targeted Radioligand Therapy (RLT) guided by the specific receptor profile of a patient's tumor—determined via baseline Radioligand Imaging (RLI)—can deliver safe, localized radiation to tumor sites, evaluating both target engagement and early efficacy signals.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Assignment (Parallel Assignment across three experimental RLT arms) **Blinding:** Open-label **Randomization:** Non-Randomized (Allocation is based on the predominantly expressed target per PET images)

3.2 Planned Enrollment & Duration Target \$N\$: 36 participants (Approx. 20 in PSMA arm, 13 in SSTR2 arm, and 3 in GRPR arm)
Number of Sites: Multi-center **Estimated Study Start:** 29-Jul-2024
Estimated Primary Completion: 27-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants, age ≥ 18 to ≤ 100 years. 2. Metastatic prostate cancer with neuroendocrine differentiation confirmed by at least one of the following: histology (small cell/neuroendocrine), expression of NEPC markers (e.g., chromogranin or synaptophysin), progression of visceral metastases in the absence of PSA progression, or molecular features (e.g., PTEN, TP53, or RB loss). 3. PSMA and/or SSTR2 and/or GRPR PET-positive participants, with at least one measurable lesion per RECIST 1.1 with moderate target expression in at least one of the 3 PET/CT scans per BICR assessment. 4. Castrate level of serum/plasma testosterone (< 50 ng/dl, or < 1.7 nmol/L) for participants with adenocarcinoma component or stable testosterone level for participants with pure neuroendocrine carcinoma. 5. Recovered to $\leq$ Grade 2 from all clinically significant toxicities related to prior therapy. 6. Participant has adequate bone marrow and organ function (as assessed by central laboratory for eligibility). 7. ECOG performance status ≤ 2 .

4.2 Exclusion Criteria 1. Previous treatment with any of the following within 6 months prior to Screening: Strontium-89, Samarium-153, Rhenium-186, Rhenium-188, Radium-223, hemi-body irradiation. 2. History of CNS metastases that are neurologically unstable, symptomatic, or receiving corticosteroids for the purpose of maintaining neurologic integrity. 3. Symptomatic cord compression or clinical or radiologic findings indicative of impending cord compression. 4. Previous PSMA, SSTR2, or GRPR targeted radioligand therapy. 5. Other concurrent cytotoxic chemotherapy, immunotherapy, radioligand therapy or investigational therapy. 6. History or current diagnosis of ECG abnormalities indicating significant risk of safety for study participants.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **[¹⁷⁷Lu]Lu-PSMA-617:** 7.4 GBq (200mCi) $\pm$ 10% (total dose of 44.4 GBq $\pm$ 10%), every 6 weeks for up to 6 cycles. * **[¹⁷⁷Lu]Lu-NeoB:** 9.25 GBq (250mCi) $\pm$ 10% (total dose of 55.5 GBq $\pm$ 10%), every 6 weeks for up to 6 cycles. * **[¹⁷⁷Lu]Lu-DOTA-TATE:** 7.4 GBq $\pm$ 10% (total dose of 44.4 GBq $\pm$ 10%), administered every 6 weeks for up to 6 cycles. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Up to 36 weeks maximum (6 cycles)

5.2 Concomitant & Prohibited Therapy Permitted: GnRH antagonists, antiemetics, and anti-nauseants. **Prohibited:** Prior targeted radiotherapy within 6 months. Specific control for proton-pump inhibitors (PPI) is required if assessing neuron-specific enolase markers among concomitant treatments.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Radioligand Imaging (e.g., [68Ga]Ga-PSMA-11, [68Ga]Ga-NeoB) to determine receptor predominance. | | **Baseline** | Day 1 | Allocation to corresponding arm, First Dose of Targeted RLT. | | **Interim** | Every 6 Weeks $\pm$ 7 Days | RLT Infusion, Safety Labs, Efficacy PET/CT Scans. | | **End of Study** | 42 days post-EOT | Final Safety Follow-Up Visit, transition to Survival/Progression follow-up. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number/extent of lesions with at least a moderate uptake of any of the Radioligand Imaging (RLI) according to visual assessment scoring scale on each corresponding targeted PET/CT scan based on blinded independent central review (BICR) assessment. 2. Percentage changes in quantitative PET/CT parameters (SUVmax, SUVmean, SUVpeak, target-positive Tumor Volume [target-TV], Total Lesion target [TL-target]) and changes in number of target-positive lesions (as per visual assessment) on each corresponding target PET/CT based on BICR. **Measurement Method:** Radioligand Imaging (RLI) using PET/CT scans and blinded independent central read.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR). * Disease Control Rate (DCR). * Duration of response (DOR). * Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs). * Radiographic disease progression. * Overall Survival (OS) follow-up.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs monitored continuously throughout the 36-week treatment phase and during the 42-day post-EOT visit. **Serious Adverse Events (SAEs):** Routine reporting within 24 hours to the sponsor pharmacovigilance team and corresponding IRBs. **Data Monitoring Committee (DMC):** Internal safety review committee governing dose-limiting toxicities and overall participant safety within Phase I provisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary target expression outcomes. Safety Set for all toxicity and AE profiles. **Sample Size Justification:** As an exploratory Phase I study, the targeted sample size of $N=36$ is based on

clinical and empirical considerations to evaluate safety signals and target engagement rather than strict statistical power assumptions. **Handling of Missing Data:** Missing data will be handled per sponsor standards for early-phase open-label oncology trials, utilizing available case analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote data monitoring to verify protocol adherence, accurate IP administration, and proper AE documentation. **Audit Trail:** Centralized read for all baseline and efficacy PET/CT imaging; secure eCRF locking mechanisms.

11. Study Identifiers & Governance

Primary ID: CAAA617H12101 **Registry Number:** NCT06379217 **Posting ID:** 61543512700935559 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** NEPC Study: An Exploratory Safety and Efficacy Study With PSMA, SSTR2 and GRPR Targeted Radioligand Therapy in Metastatic Neuroendocrine Prostate Cancer **Official Regulatory Title:** A Phase I, Open-label, Multi-center Exploratory Safety and Efficacy Study With PSMA, SSTR2 and GRPR Targeted Radioligand Therapy in Metastatic Neuroendocrine Prostate Cancer

12. Clinical Framework (mNEPC Specific)

Primary Condition: Metastatic Neuroendocrine Prostate Cancer (mNEPC) **Diagnosis Threshold:** Histological confirmation or specified biomarker expression (e.g., Chromogranin A > 5x normal limit) **Medical Methodology:** Targeted Radioligand Therapy (RLT) matching receptor predominance **Optimization Target:** Target engagement (PSMA, SSTR2, GRPR) and radiographic tumor response

13. Study Procedures & Methodology

Management Pathway: Imaging-guided RLT allocation (baseline RLI determines the specific RLT agent based on BICR) **Education Protocol (HCP):** Site radiation safety protocols, RLT dosing logistics, and imaging acquisition standardizations. **Education Protocol (Patient):** Radiation safety precautions post-infusion and symptom management instructions. **Quality Audit Mechanism:** Blinded independent central review (BICR) to dictate treatment arm assignment.

14. Observation & Follow-Up Schedule

Total Duration: Up to 36 weeks of treatment, followed by survival/progression tracking. **Onsite Visit Cadence:** Every 6 weeks during the treatment phase (Days 1 of Cycles 1-6) plus a 42-day EOT visit. **Remote Monitoring Frequency:** Periodic survival follow-ups post-progression until death or withdrawal. **Checklist Review Interval:** Routine clinical assessments prior to each 6-week RLT infusion.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				